

MAYANK GARG

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EDUCATION

Thapar Institute of Engineering & Technology (Current CGPA: 9.42)

Patiala, Punjab, India

Bachelor of Technology in Computer Science Engineering

July 2025 – May 2028

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Web Development, Database Management Systems

Diploma in Computer Science Engineering

Graduated 2025

Thapar Polytechnic College

- Completed comprehensive program covering programming fundamentals, web technologies, and software development practices

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript, HTML5, CSS3

Frameworks & Libraries: React.js, Node.js, Express.js, Tailwind CSS, Bootstrap, Scikit-learn

Developer Tools: Git, GitHub, VS Code, Postman

Databases: MongoDB

Core Competencies: Full-Stack Development, Machine Learning, Object-Oriented Programming in C++ Responsive Web Design, Data Structures & Algorithms, Problem Solving

PROJECTS

AQI Prediction System | Python, Scikit-learn, Pandas, NumPy, Matplotlib | 🐙

2025

- Built a machine learning project to predict Air Quality Index(AQI) using environmental data such as PM2.5, CO and many other pollutants.
- Cleaned and prepared real-world datasets by handling missing values, removing outliers, and creating useful features to improve model performance
- Trained and tested a Random Forest model to generate accurate AQI predictions and evaluated results using standard performance metrics

Virtual Keyboard with Hand Gesture Control | Python, OpenCV, MediaPipe, PyAutoGUI | 🐙

2024

- Engineered touchless typing system using computer vision and hand gesture recognition, enabling users to type without physical keyboard contact through webcam-based finger tracking
- Implemented distance-based gesture detection algorithm using MediaPipe hand landmark tracking, triggering key presses when index and middle finger proximity falls below configurable threshold
- Integrated OpenCV for real-time video processing and PyAutoGUI for keystroke simulation, achieving 95% gesture recognition accuracy with sub-100ms latency

Imagify - AI Image Generation Platform | React.js, Node.js, Express.js, MongoDB, Stripe API, AI Image API | 🐙

2024

- Developed full-stack AI-powered image generation platform with complete MERN authentication system, enabling users to create custom images from text prompts using advanced AI models
- Integrated third-party AI image generation API with prompt optimization engine, delivering high-quality images with automatic download functionality and gallery management
- Designed modern, responsive UI with intuitive prompt interface, credit balance indicators, pricing plans showcase, and seamless user experience across all devices

ACHIEVEMENTS & EXTRACURRICULAR

Hackathon Achievement: Secured 3rd place in Internal Hackathon during Diploma (2nd year) for developing innovative web-based solution

Competitive Programming: Actively solving problems on LeetCode and CodeChef to strengthen data structures and algorithms knowledge

Self-Learning: Completed multiple online tutorials and built 3+ full-stack projects and 1 Machine Learning independently within 1 year of learning MERN stack and Machine Learning